

Econ 139: Labor Economics

Technology: Automation

David Arnold



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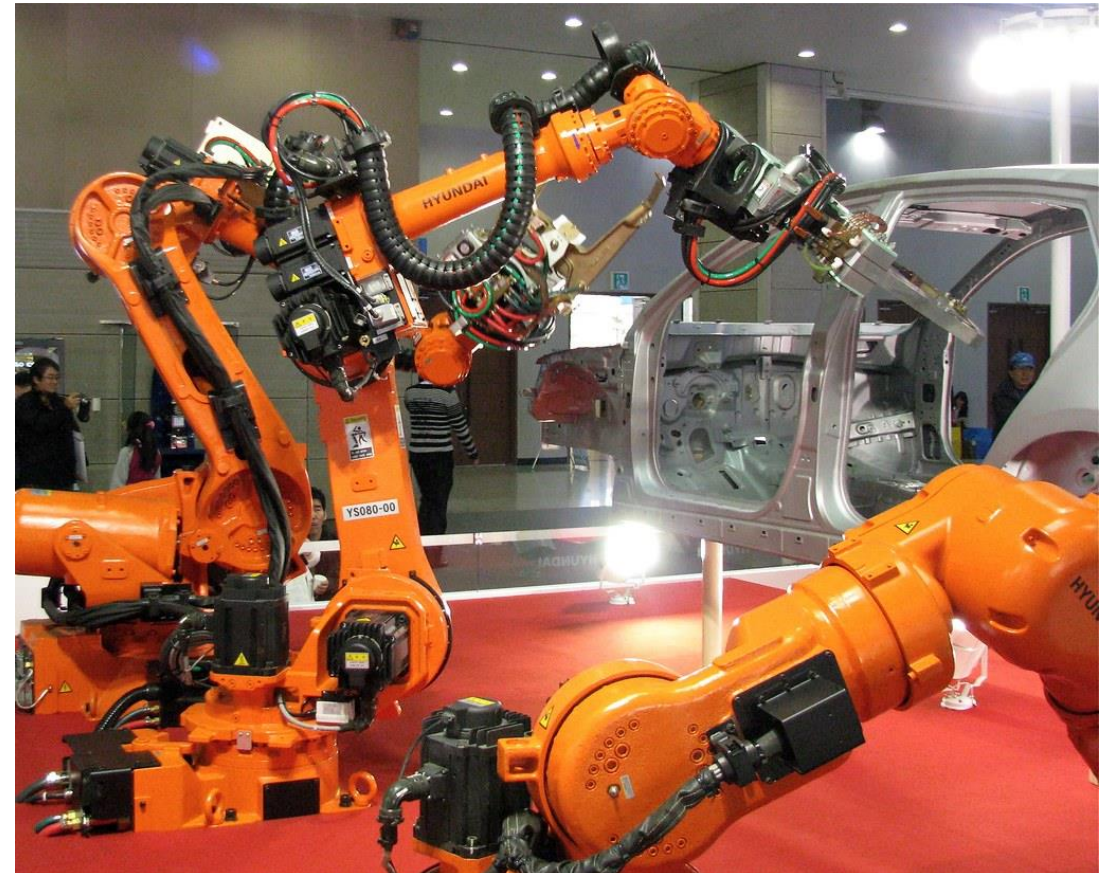


Review

- Last time: task-based model predicts which occupations will see relative declines in employment
 - Patterns in data consistent with model in which computers/technology replaced middle-income routine jobs starting in the 1980s
- Today we will focus one specific technology: industrial robots
 - This will allow us to provide direct evidence that technology adoption *causes* changes in employment and wages

Industrial Robots

- Acemoglu and Restrepo (2020) study one particular form of automation: industrial robots
- Industrial robots are defined as: “an automatically controlled, reprogrammable, and multipurpose machine”
- Robotics technology advanced significantly in the 1990s and 2000s leading to large increases in the use of industrial robots



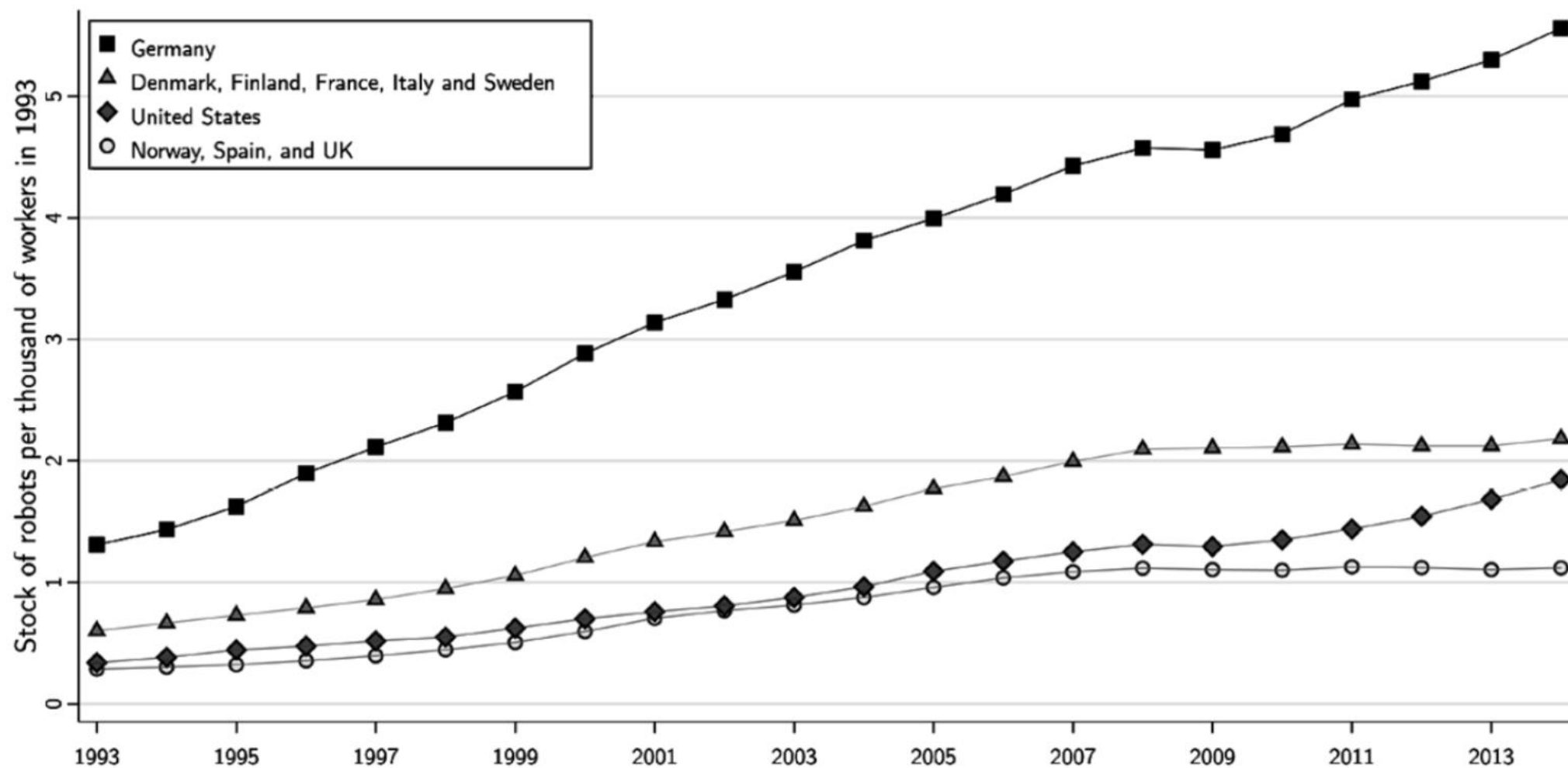


FIG. 1.—Industrial robots per thousand workers in the United States and Europe.

Measuring Exposure to Robots

- Acemoglu and Restrepo (2020) compute a measure of exposure to industrial robots for every commuting zone in the US
 - A commuting zone is a group of counties that tend to be connected economically
- Basic idea: they know what industries are adopting robots. They also know where industries are located
- So places with high exposure to robots are places with high employment in industries that adopted industrial robots

Example Measure

- Imagine we have two commuting zones, CZ A and CZ B
- CZ A has 50-50 split of employment into auto and services
- CZ B has 10 percent of employment in auto and 90 percent in services
- Based on data from IFR, we know robots grew by 10 robots per thousand workers in automotive and services adopted no robots

Example Measure

- How many robots per worker do we expect in CZ A?

$$Exposure = \underbrace{10 + 0.5}_{\text{Change in Auto}} + \underbrace{0 + 0.5}_{\text{Change in services}} = 5$$

- Intuitively, I expect 5 more robots per thousand workers in CZ A

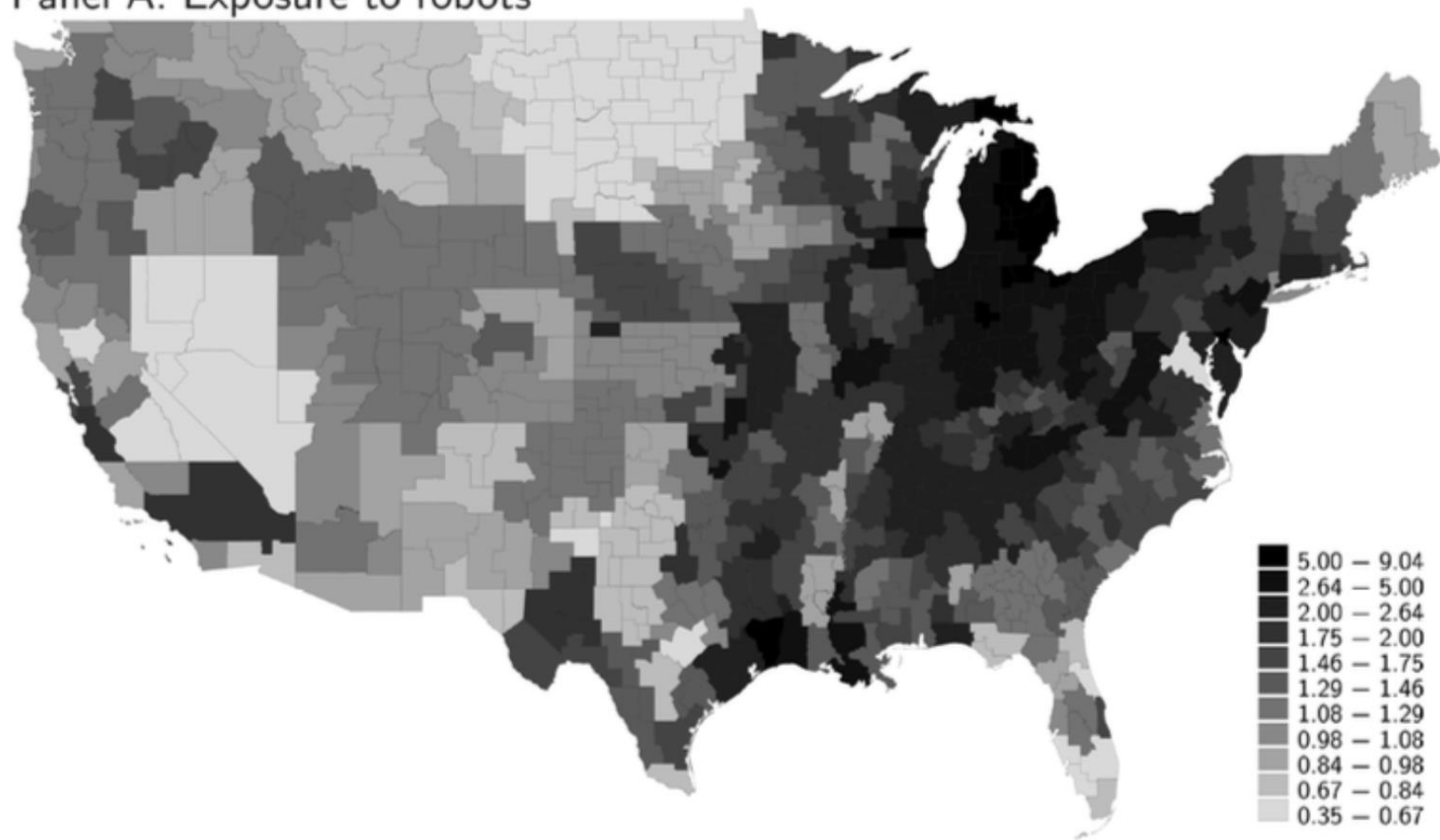
Example Measure

- How many robots per worker do we expect in CZ B?

$$Exposure = \underbrace{10 + 0.1}_{\text{Change in Auto}} + \underbrace{0 + 0.9}_{\text{Change in services}} = 1$$

- Intuitively, I expect 1 more robot per thousand workers in CZ B

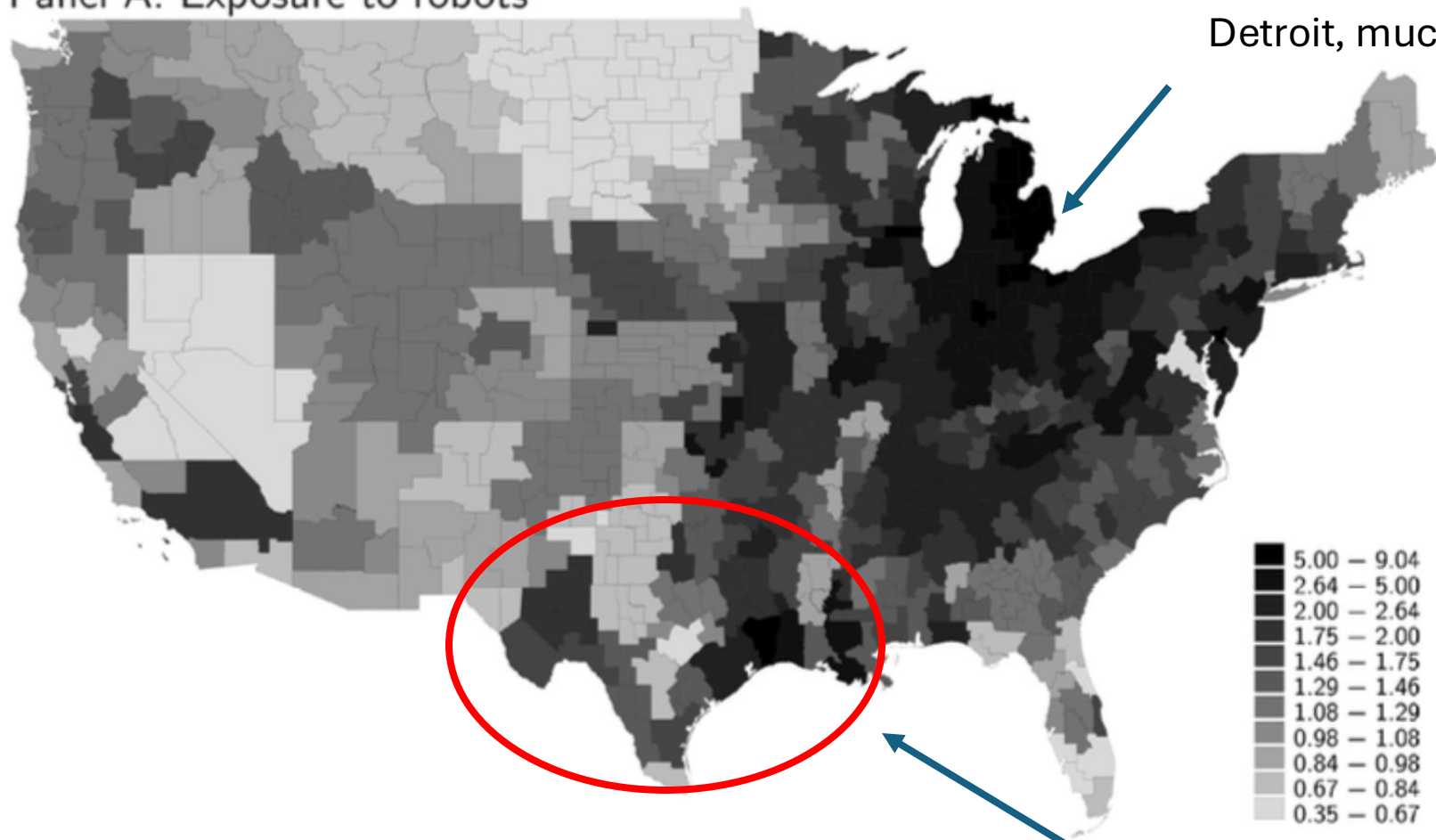
Panel A. Exposure to robots



[Exposure to Industrial Robots by Commuting Zone](#)

Panel A. Exposure to robots

Manufacturing areas, such as
Detroit, much more exposed

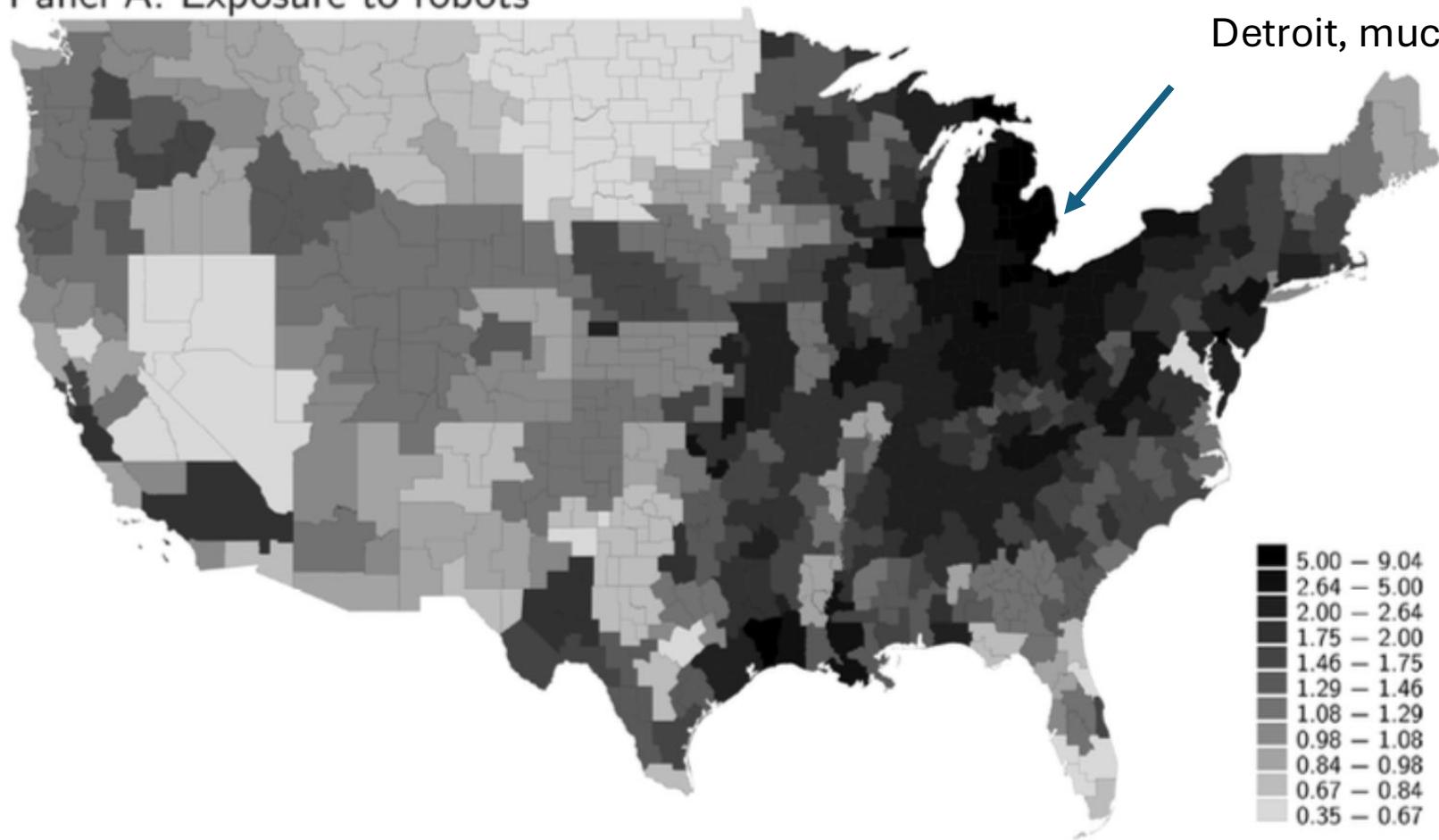


[Exposure to Industrial Robots by Commuting Zone](#)

Even within some states, lots of variation, though

Panel A. Exposure to robots

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[Exposure to Industrial Robots by Commuting Zone](#)

Employment to Population Ratio

- Main outcome is employment to Population Ratio
 - Number of employed individuals per entire population
- To understand the design, let's again return to CZ A and CZ B

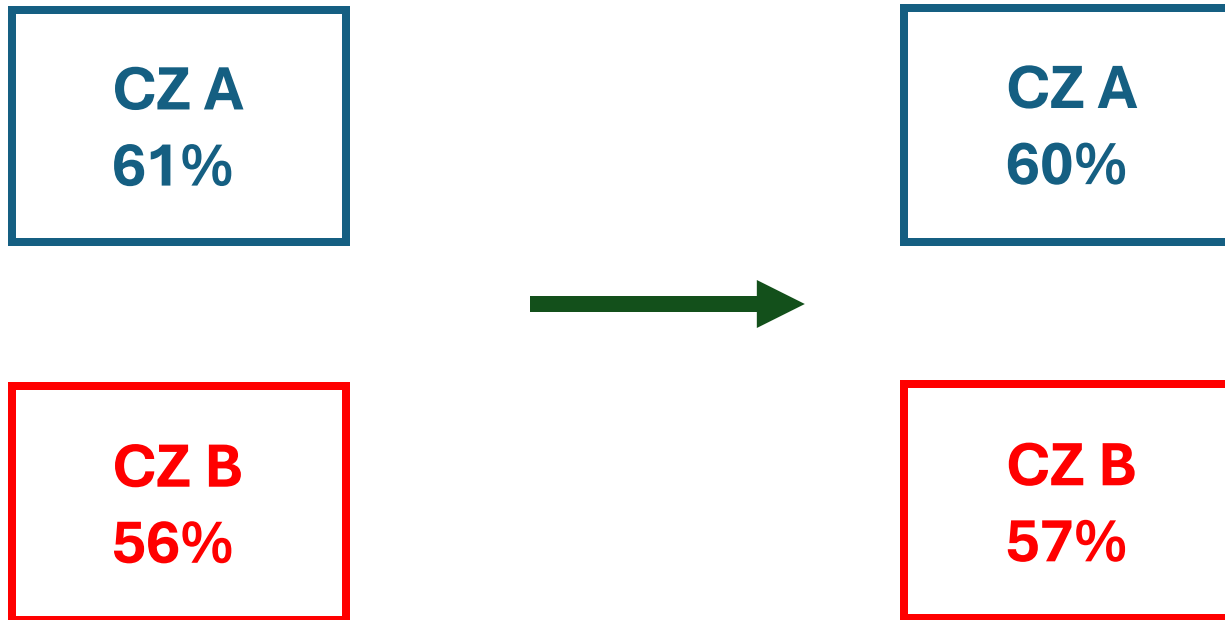
Employment to Population Ratio

CZ A
61%

CZ B
56%

1993 Employment to Population Rates

Employment to Population Ratio



1993 Employment to Population Rates

2007 Employment to Population Rates

Employment to Population Ratio

CZ A
61%

CZ A
60%

CZ A
-1%



CZ B
56%

CZ B
57%

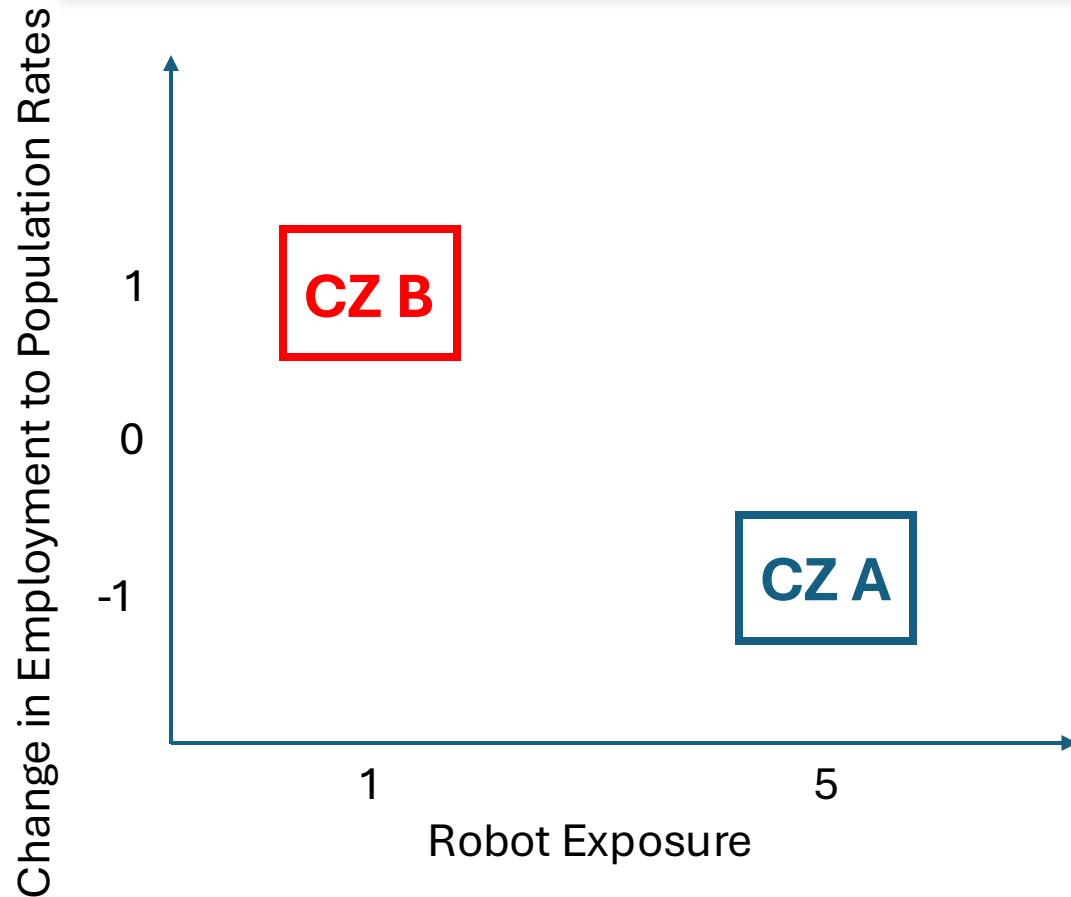
CZ B
+1%

1993 Employment to Population Rates

2007 Employment to Population Rates

Change

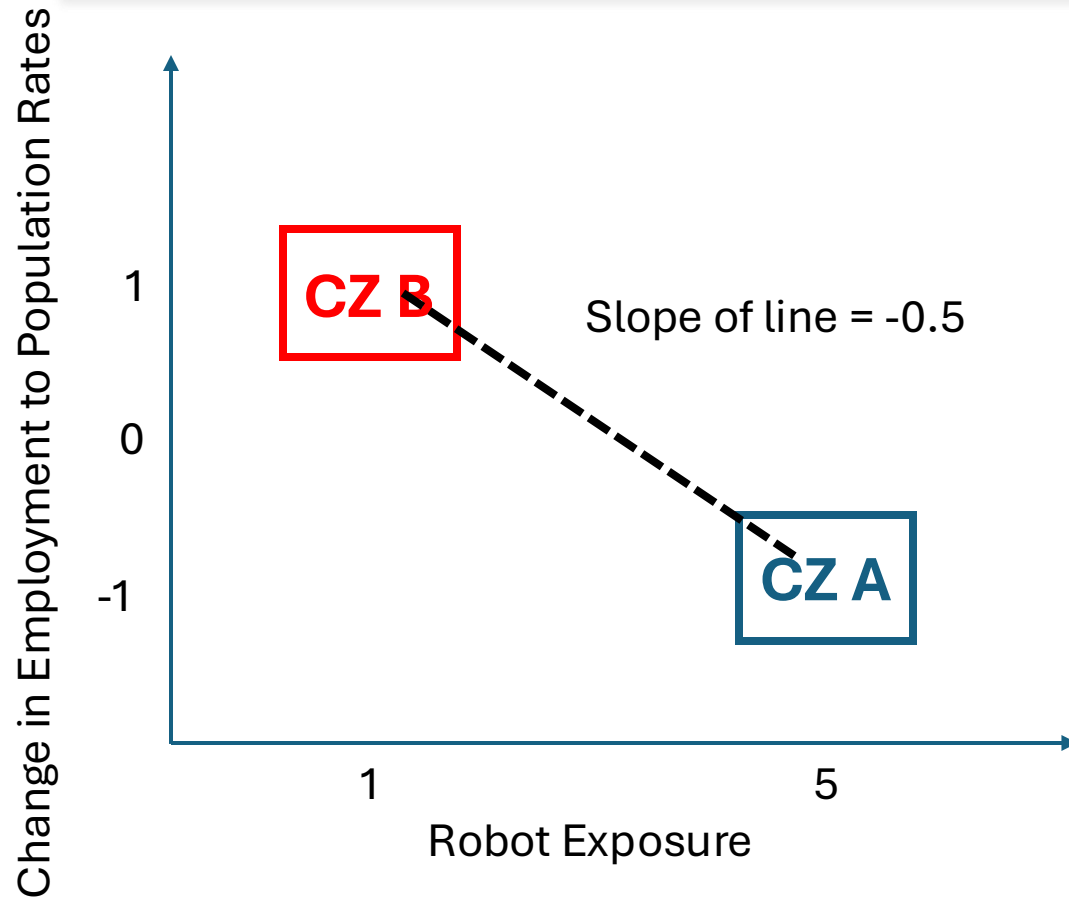
Employment to Population Ratio



Interpretation: 4 more robots per thousand workers decreases employment to population rate by 2 percentage points

In terms of a single robot: 1 more robot per thousand workers decreases employment to population rate by 0.5 percentage points

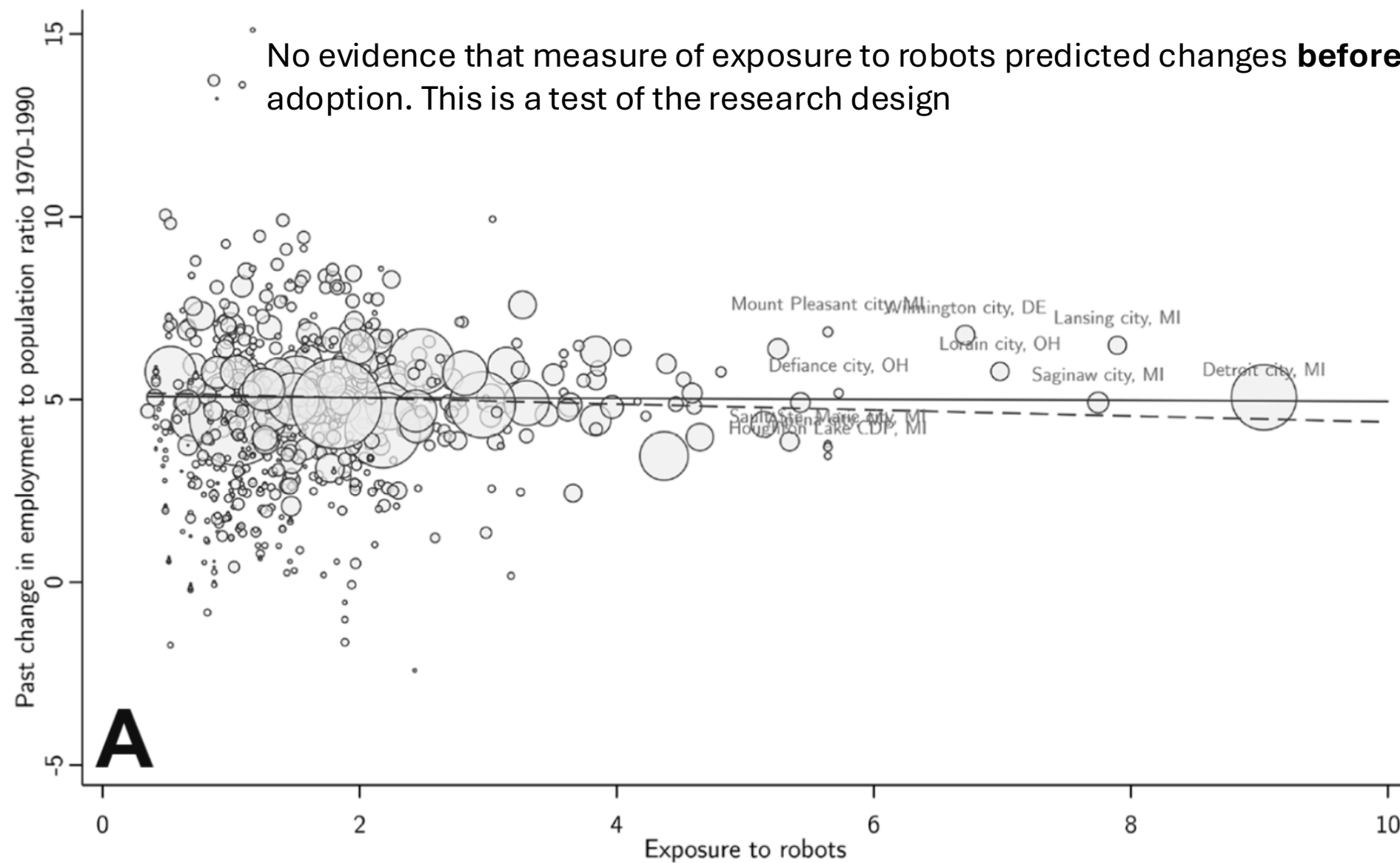
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No evidence that measure of exposure to robots predicted changes **before** robot adoption. This is a test of the research design





Places where we predicted robots would be adopted (based on industries) saw declines in employment relative to places that we predicted fewer robots would be adopted. Can we conclude that in aggregate robots lowered employment?

General Equilibrium Effects

- Imagine car manufacturers in Detroit reduce labor considerably, but start selling cars for less
- This price reduction benefits people both inside Detroit, but also **outside of Detroit**
- What if other sectors expand as individuals have more money to spend on items besides cars
- Then robots decreased employment in automotive industries, but increased employment in other sectors

How to Answer This?

- This is a fundamentally difficult question with no obvious solution
- One common approach is to build a mathematical model of the economy
- This can be used to translate these estimates across commuting zones into aggregate impacts by modeling how commuting zones may trade goods, leading to output increases and employment changes
- However, it relies on further assumptions that can be hard to verify. This is the price of using models. The benefit is you can answer hard questions

Overall Takeaway

- Acemoglu and Restrepo (2020) use their model to compute that one more robot per thousand workers causes **aggregate demand** to fall by 0.2 percentage points
- Putting this number into aggregate jobs lost due to robots (up until the author's data): 400,000 jobs

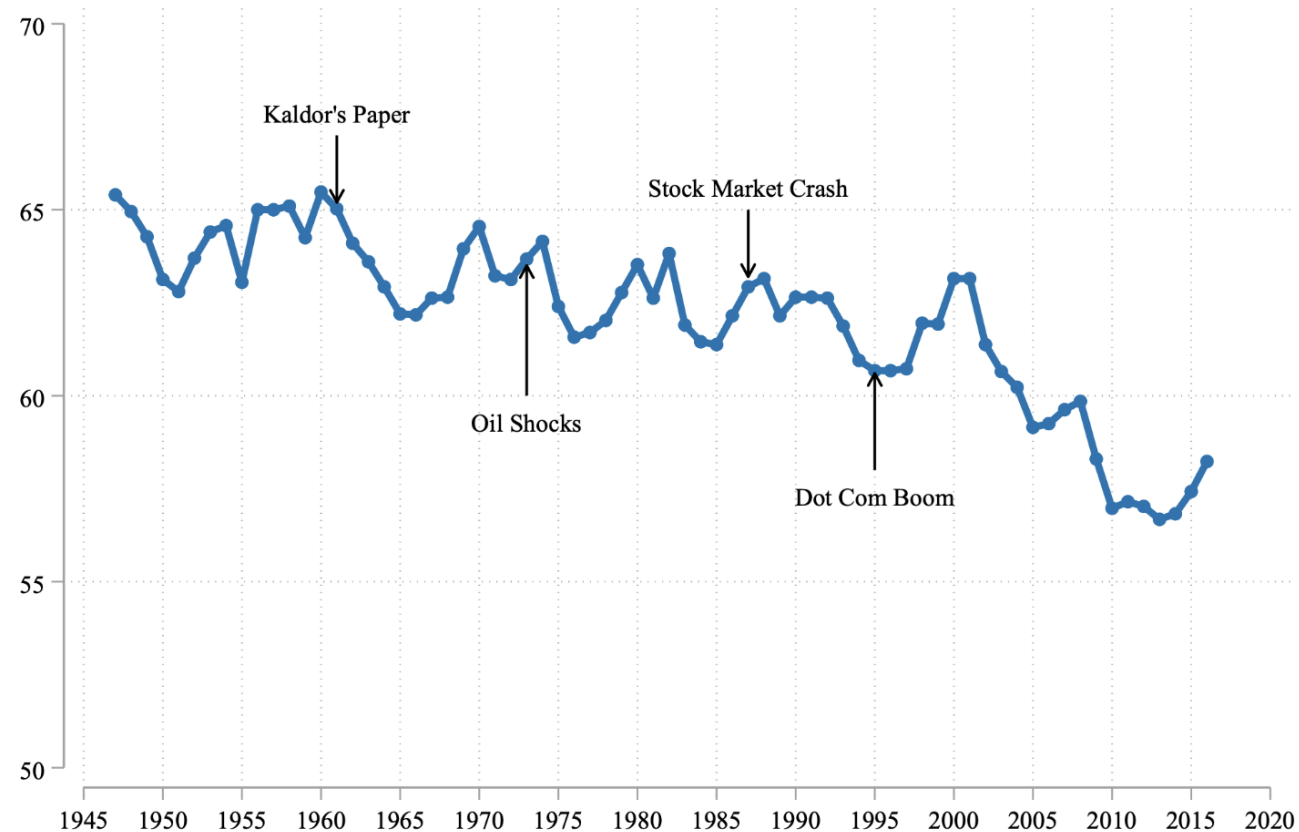
The Labor Share

- To remind you, the labor share of the economy is given by

$$\text{Labor Share} = \frac{WL}{GDP}$$

- Acemoglu and Restrepo (2020) found automation due to industrial robots decreased L , and while we did not focus on this, they also decrease W (wages)
- In a follow-up paper, Acemoglu and Restrepo (2022) explore how much of the fall in the labor share can be attributed to technology adoption, including adoption of robots but also specialized software and dedicated machinery

Reminder: The Fall in the Labor Share



Automation Technologies

- In Acemoglu and Restrepo (2022) the authors expand automation technologies to include software and dedicated machinery
 - Dedicated machinery is a broader notion of machine than industrial robot
 - Does not need to be programmable like an industrial robot
 - Can be dedicated to a single task, with or without help of a human
- For each industry, they compute the change in adoption of these technologies and correlate this with the change in the labor share

Change in Specialized Software and Dedicated Machinery

- Take an industry producing 10 billion in value added in 1987 and using 1 billion worth of specialized software and dedicated machinery then the
 - Ratio of technology costs to value added would be 0.1.

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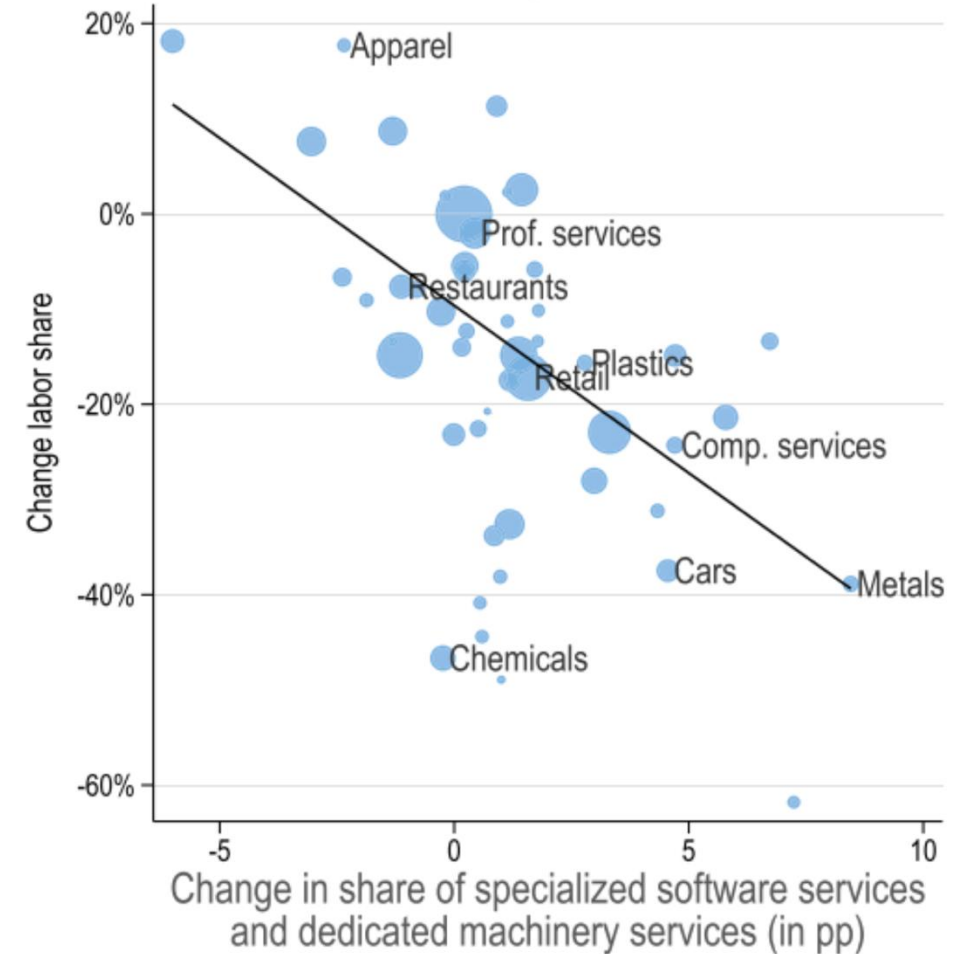
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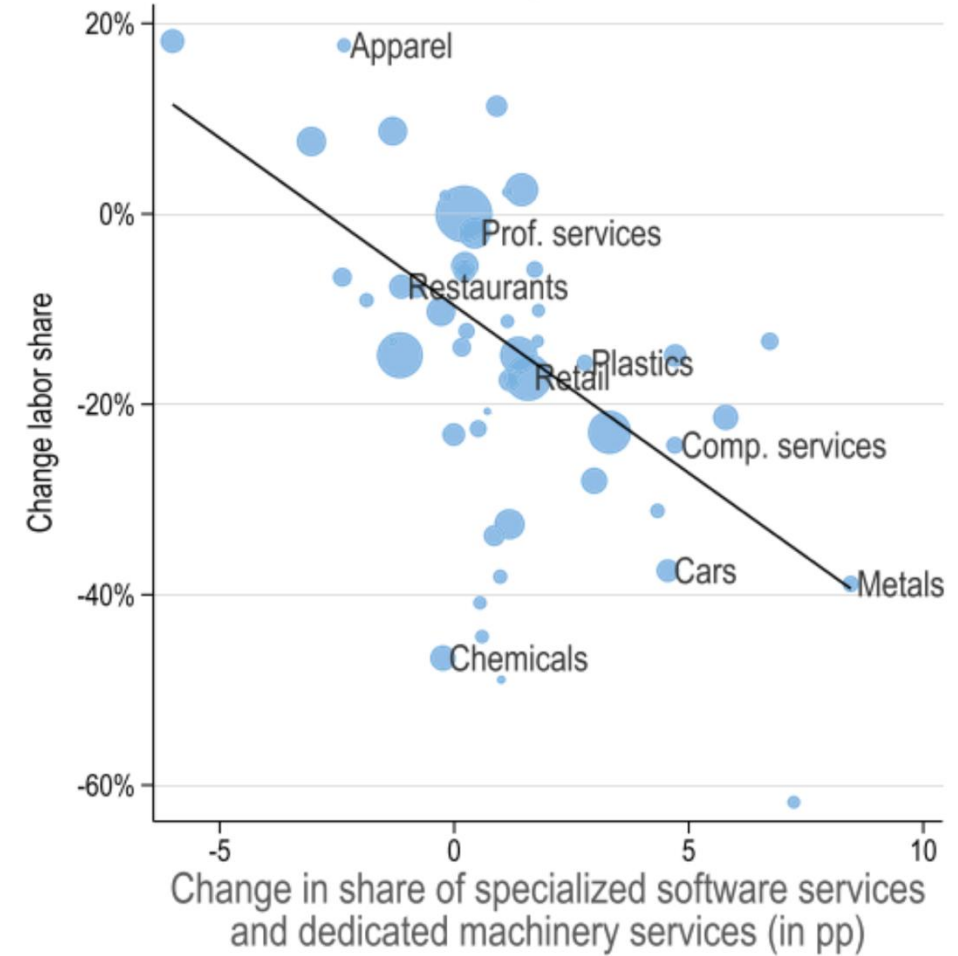
B. Labor share vs. specialized software and dedicated machinery, 1987-2016



Change in Specialized Software and Dedicated Machinery

- Adding industrial robots to the mix, we can explain 45% of the variation in the labor share changes across industries using technology
- This is (maybe) surprising! Lots of things were happening between 1987-2016, but this evidence suggests that technology is a primary driver of the fall in the labor share across industries

B. Labor share vs. specialized software and dedicated machinery, 1987-2016



Inequality

- Next let's link this to inequality
- We will do this in two steps:
 - Humlum (2021) studies firm-level adoption of robots and how it impacts different type of workers
 - Acemoglu and Restrepo (2022) study how groups exposed to automation technologies (i.e. generally less-educated workers) earnings change relative to groups less exposed (i.e. generally more-educated workers)

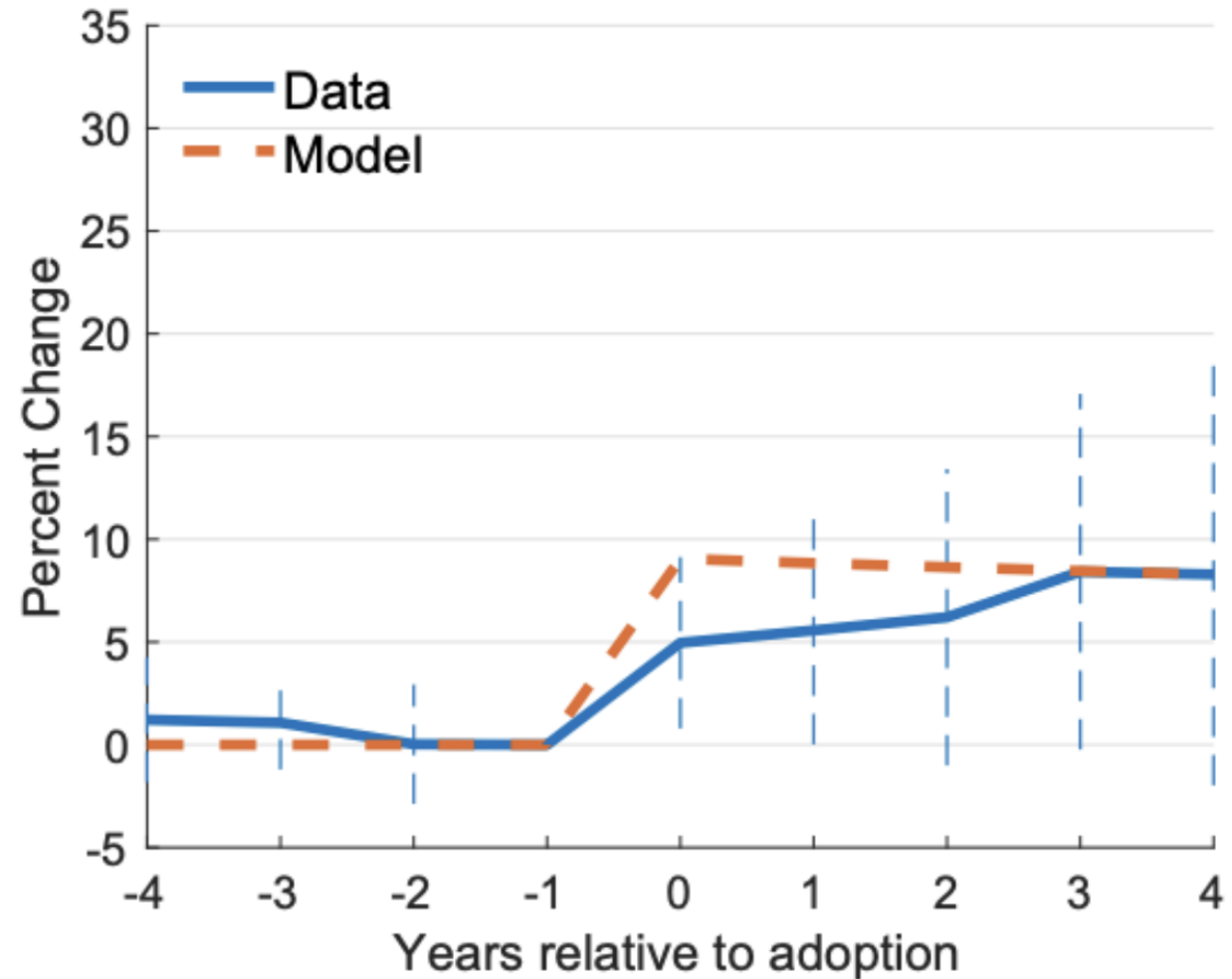
Firm-Level Evidence

- Humlum (2021) uses impressive data in Denmark that includes
 - At the firm-level, the date they buy a robot
 - For each worker at the firm, their occupation and earnings
 - Includes information on the **universe** of workers in Denmark
- Detail of data allows author to study changes at firms that adopt robots relative to firms that don't adopt robots

Surprising Results: Firms that adopt robots spend **more** on workers

- Wage bill is equal to total amount spent on labor
- Blue line is different between adopters and non-adopters
- For example, value of 5 indicates wage bill for adopters grows by 5 percent relative to wage bill for non-adopters

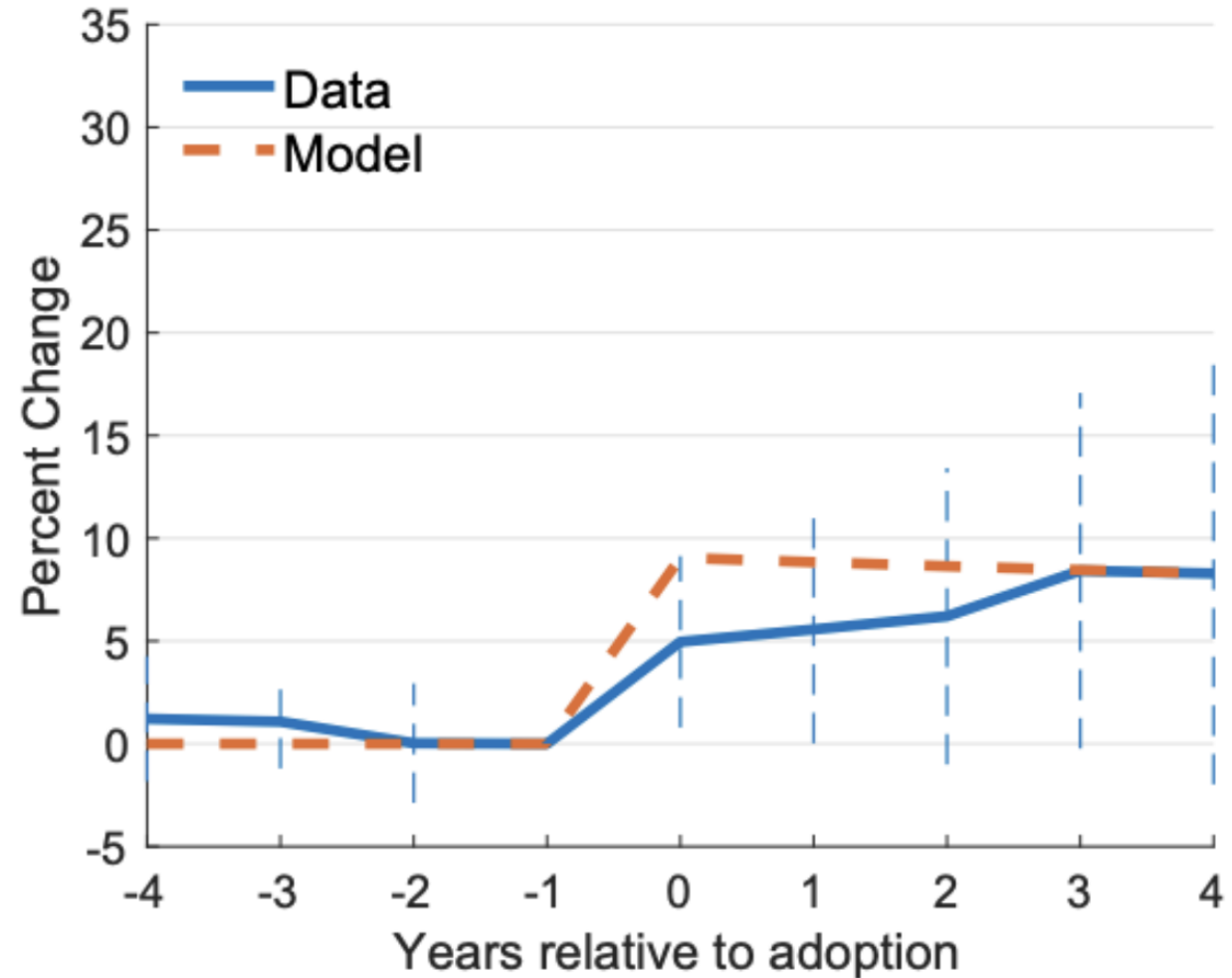
(b) Wage Bill



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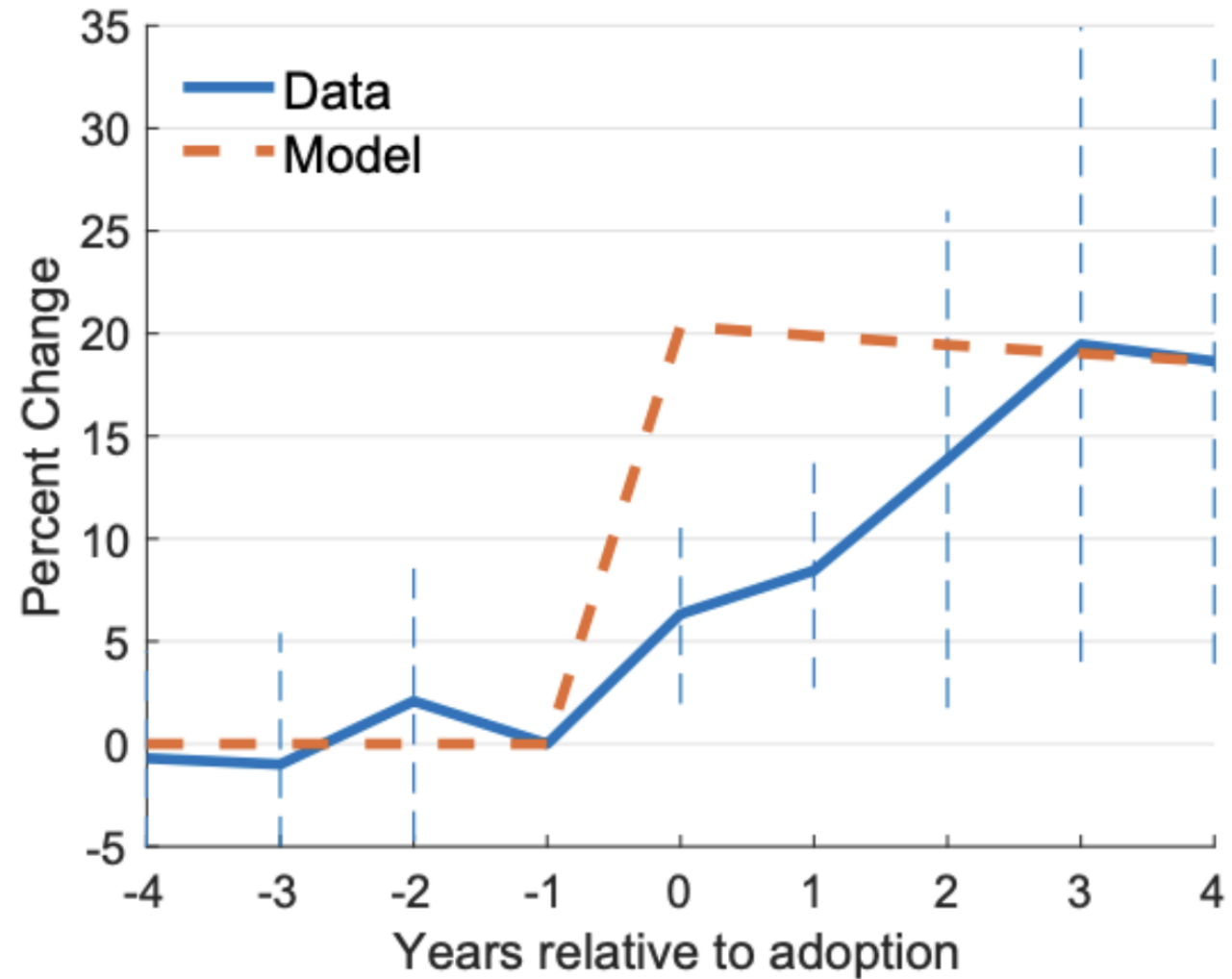
- How can we reconcile this finding with Acemoglu and Restrepo (2020)
- Well, robot adopters are spending more on workers because they are expanding significantly!



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(a) Sales



Competitors

- Competitors of robot adopters should be potentially worried
- Acemoglu, Lelarge and Restrepo study similar context and find competitors to robot adopters see large declines in employment
- Therefore, overall impact of robot on employment in the market is negative

(a) Sales

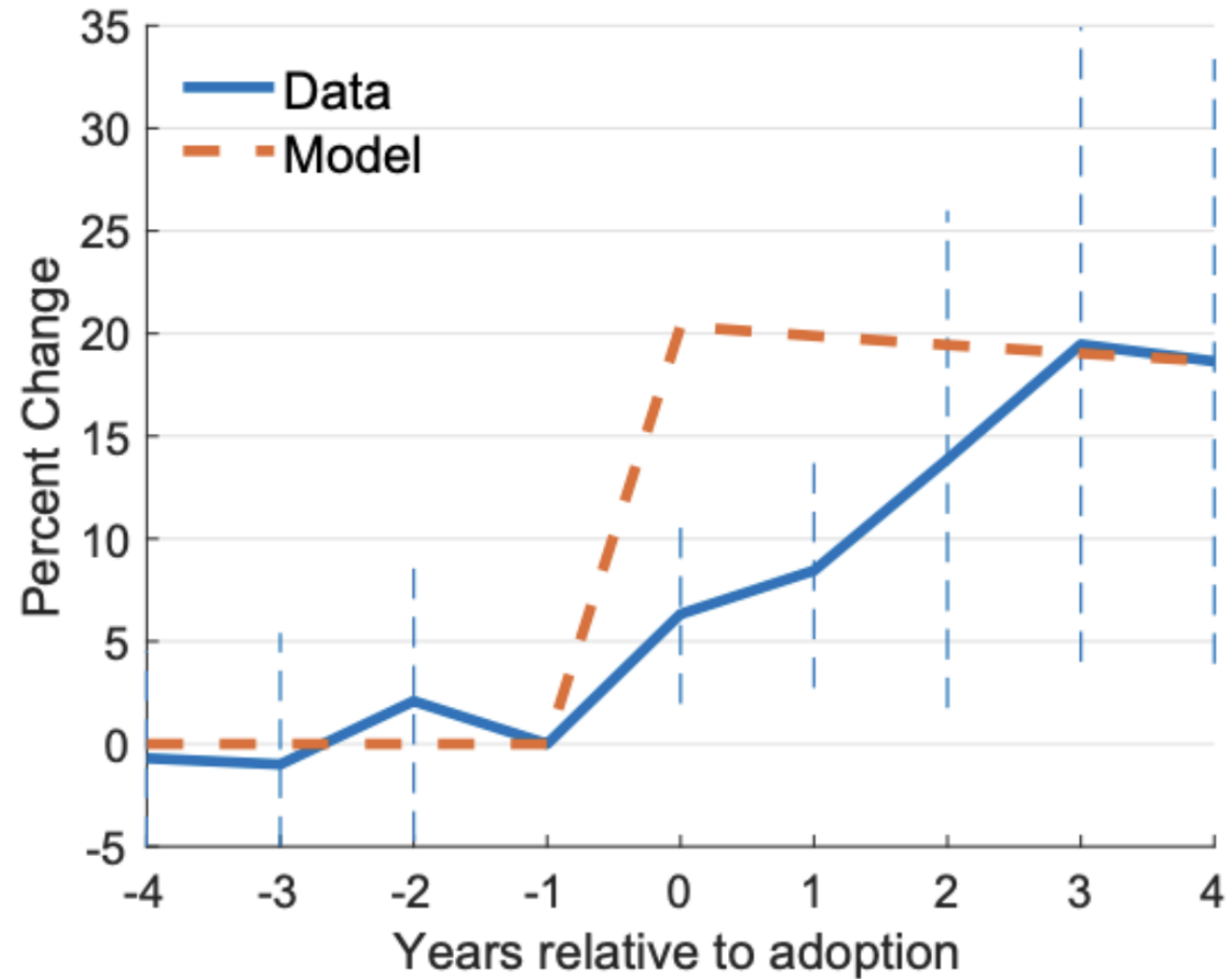
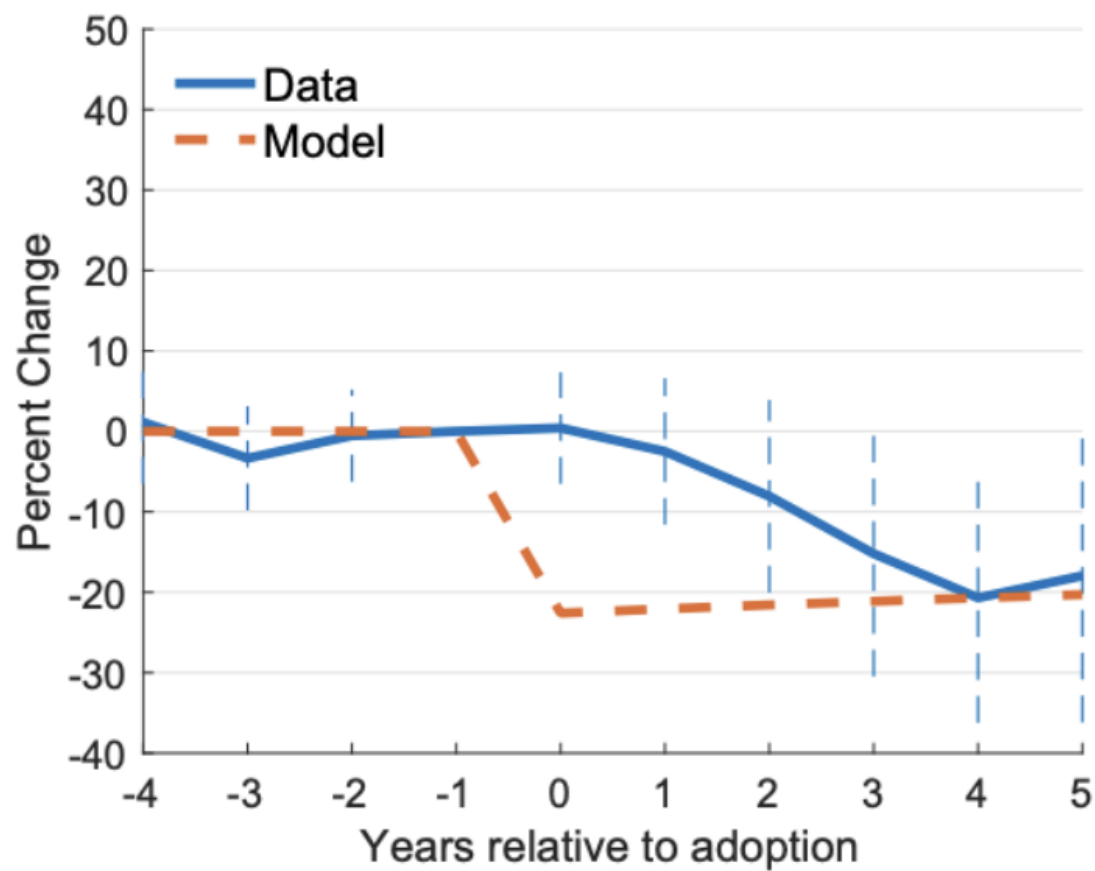


Figure 2: Firm Wage Bills Around Robot Adoption (Matching Diff-in-Diff)

(a) Production Workers



(b) Tech Workers

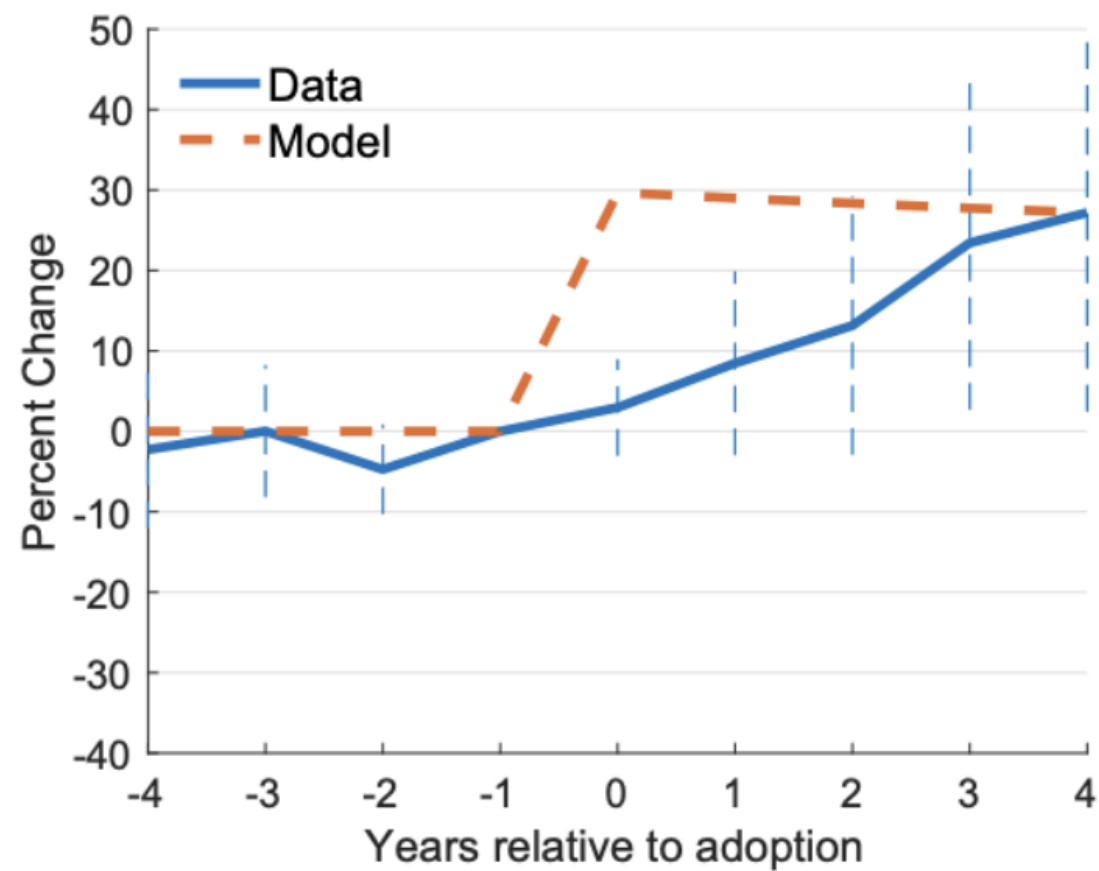
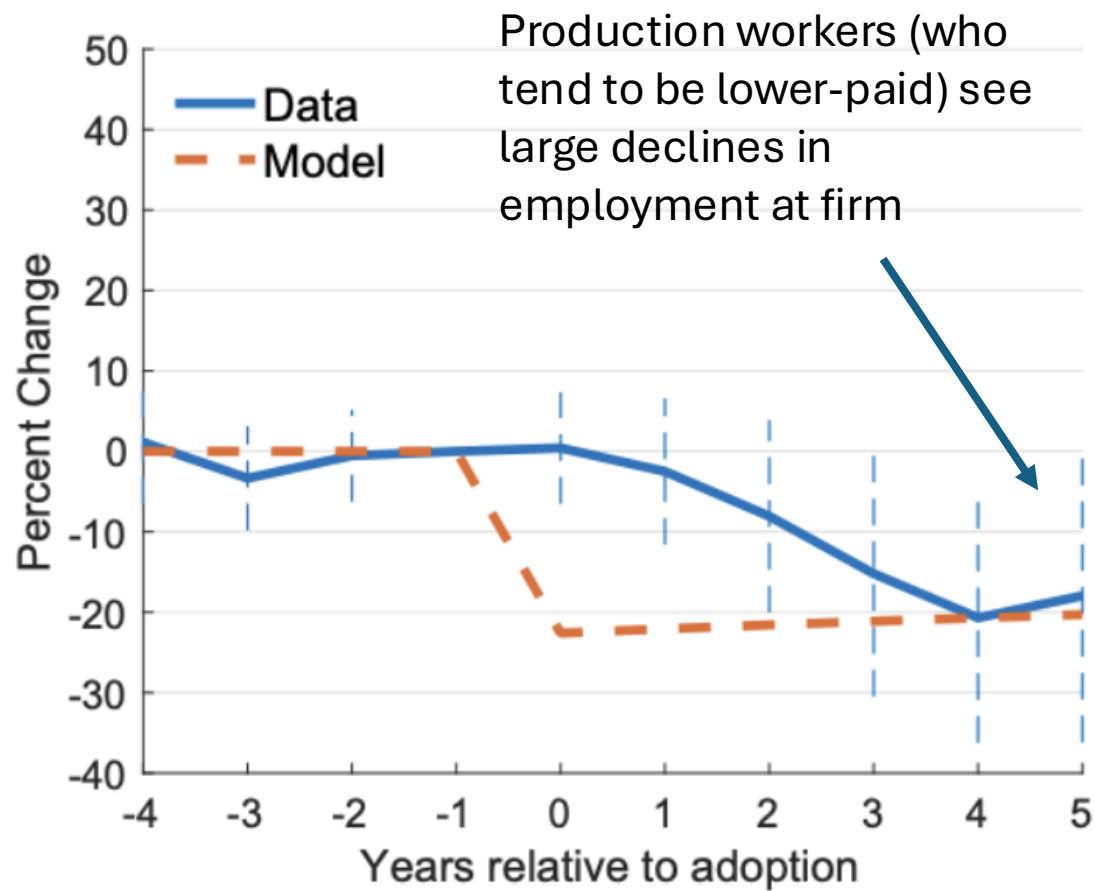


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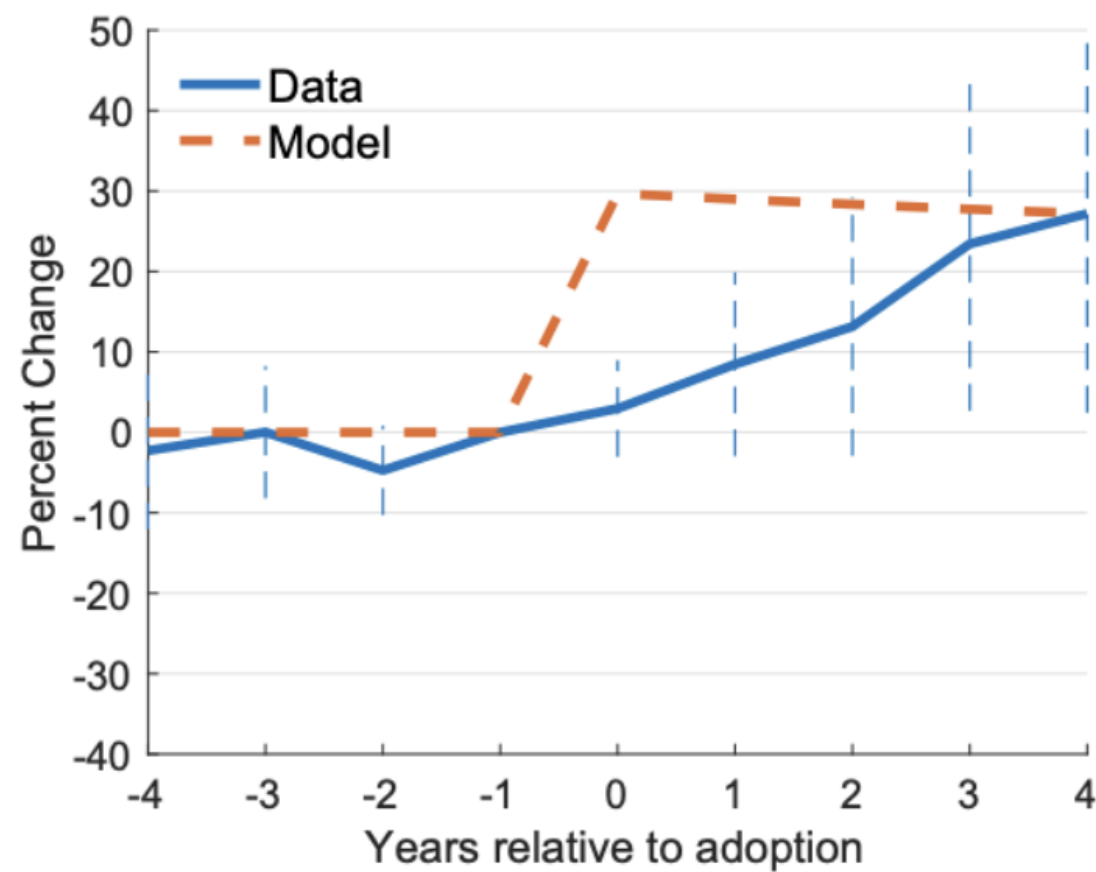
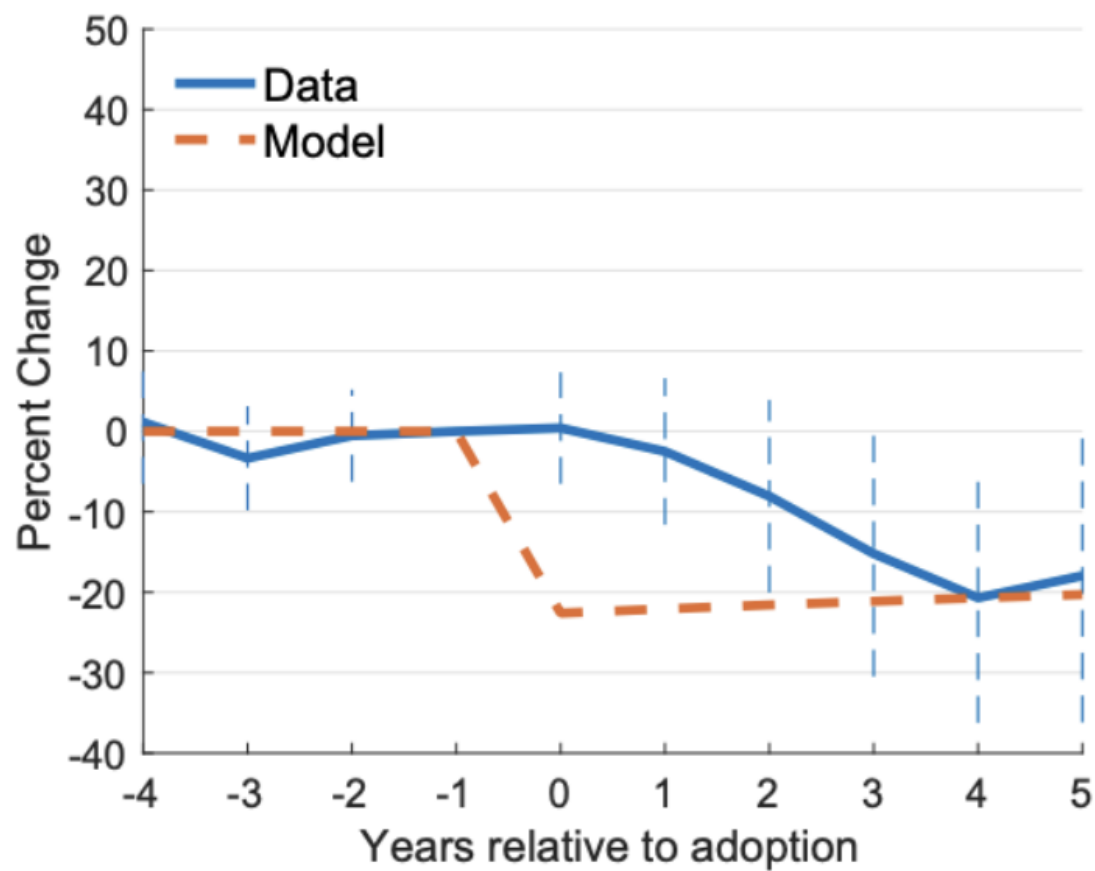
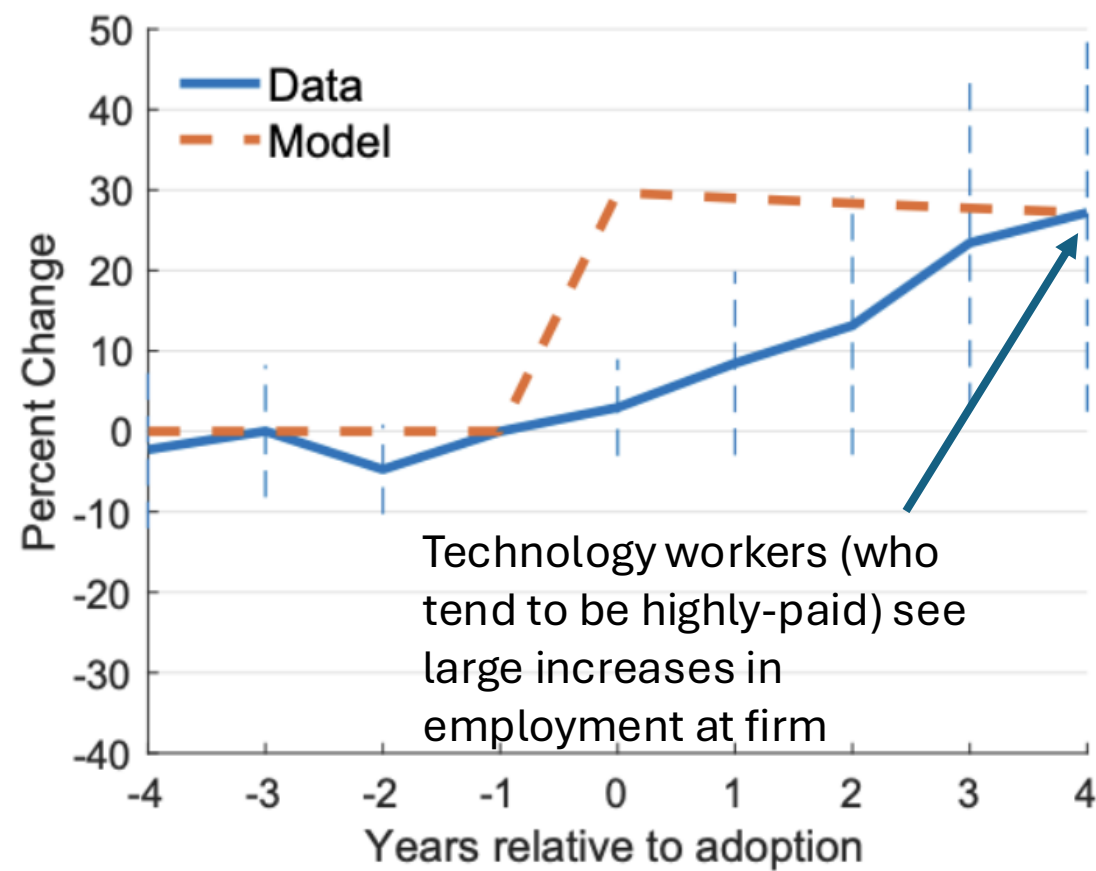


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(a) Production Workers



(b) Tech Workers



Overall Takeaway

- Humlum (2021) finds that robot adoption leads to increases in inequality
 - Production workers wages are 5.4% less in aggregate due to robot adoption
- This is clear and compelling evidence that robot adoption leads to rises in inequality, but again, this paper focuses only on robots
- To understand total impact, we can return to Acemoglu and Restrepo (2022), who explore the impact of robots + specialized software + dedicated machinery on inequality (in addition to the labor share, which we have already discussed)

Overall Takeaway

- Each bubble is a specific demographic group based on education, age, race/ethnicity, and nativity
- Postgraduate (higher earners) saw enormous increases in hourly wages between 1980-2016
- High school dropouts/high school graduates saw decreases in hourly wages between 1980-2016
- Acemoglu and Restrepo (2022) find that their measure of automation can explain **45% of the rise in inequality between these demographic groups in this period**

